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用系统思维变革食物系统

议论文 · 科学与社会 · 全文 407 词 · 逐句精读

¹ “What would the world be like if there were no hunger?”

“如果没有饥饿，世界会是什么样子？”

hunger n. 饥饿

² It's a question that Professor Crystal would ask her students.

这是克利斯特尔教授常会问学生的一个问题。

question n. 问题

professor n. 教授

³ They **found it hard to** answer, she wrote later, because **imagining** something that isn't **part of** real life—and learning how to make it real—is a **rare skill**.

她后来写道，学生们觉得这个问题很难回答，因为想象现实生活中并不存在的事物，并学会如何把它变成现实，是一种难得的本领。

find it hard to 觉得做……很难

imagine v. 想象；设想

part of的一部分

rare adj. 罕见的；难得的

skill n. 技能；本领

⁴ It is taught to **artists** and **engineers**, but **much less often** to **scientists**.

艺术家和工程师会接受这种本领的训练，而科学家接受这种训练的机会则少得多。

artist n. 艺术家

engineer n. 工程师

much less often 少得多；频率低得多

scientist n. 科学家

⁵ Crystal **set out to** change that, and helped to create a **global movement**.

克利斯特尔着手改变这一状况，并帮助发起了一场全球性运动。

set out to 着手（做）；开始（做）

global adj. 全球的

movement n. 运动

⁶ The **result**—an **approach** **known as** **systems thinking**—is now **seen as** **essential** in **meeting** global **challenges**.

其成果是一种名为“系统思维”的方法，如今被视为应对全球性挑战必不可少的手段。

result n. 结果；成果

approach n. 方法

known as 被称为；名为

systems thinking 系统思维

be seen as 被视为

essential adj. 必不可少的；极其重要的

meet v. 应对；满足（要求）

challenge n. 挑战

- 7 Systems thinking is **crucial** to **achieving targets** **such as** zero hunger and better **nutrition** because it **requires considering** **the way in which** food is **produced, processed, delivered** and **consumed**, and looking at how those things **intersect** (交叉) with human health, the **environment, economics** and **society**.

系统思维对于实现“零饥饿”和“改善营养”等目标至关重要，因为它要求考虑食物生产、加工、运送和消费的方式，并审视这些环节如何与人类健康、环境、经济和社会相互交叉。

crucial adj. 至关重要的	achieve v. 实现；达到	target n. 目标	such as 例如	nutrition n. 营养
require v. 要求；需要	consider v. 考虑	the way in which的方式		produce v. 生产
process v. 加工	deliver v. 运送；配送	consume v. 消费；食用	intersect v. 交叉；相交	
environment n. 环境	economics n. 经济；经济学	society n. 社会		

8 According to systems thinking, changing the food system—or any other network—requires three things to happen.

按照系统思维，要改变食物系统或其他任何网络，需要做到三件事。

according to 根据；按照

system n. 系统；体系

network n. 网络

require v. 需要；要求

- 9 First, **researchers** need to **identify** all the **players** in that system; second, they must **work out** how the players **relate to** each other; and third, they need to understand and **quantify** the **impact** of those **relationships** on each other and on those **outside** the system.

第一，研究人员需要找出该系统中的所有参与者；第二，他们必须弄清这些参与者彼此之间有何关联；第三，他们需要理解并量化这些关系对彼此以及对系统外各方的影响。

researcher n. 研究人员

identify v. 识别；找出

player n. 参与者

work out 弄清；想出

relate to 与……有关联

quantify v. 量化

impact n. 影响

relationship n. 关系

outside prep. 在……之外

10 Take nutrition.

以营养为例。

take v. 以……为例

nutrition n. 营养

- ¹¹ In the **latest** UN report on global **food security**, **the number of undernourished** (营养不良) people in the world has been rising, **despite** great **advances** in nutrition science.

根据联合国最新发布的全球粮食安全报告，尽管营养科学取得了巨大进步，世界上营养不良的人数却一直在上升。

latest adj. 最新的

food security 粮食安全

the number of的数量

undernourished adj. 营养不良的

despite prep. 尽管

advance n. 进步；进展

12 **Tracking** of 150 **biochemicals** in food has been important in **revealing** the relationships between **calories**, sugar, **fat** and the **occurrence** of **common** **diseases**.

追踪食物中的150种生化物质，对于揭示热量、糖、脂肪与常见疾病发生之间的关系一直很重要。

tracking n. 追踪；跟踪

biochemical n. 生化物质

reveal v. 揭示

calorie n. 卡路里；热量

fat n. 脂肪

occurrence n. 发生；出现

common adj. 常见的

disease n. 疾病

13 But using **machine learning** and **artificial intelligence**, some scientists **propose** that human **diets** **consist of** **at least** 26,000 biochemicals—and that **the vast majority** are not known.

但一些科学家借助机器学习和人工智能提出，人类饮食由至少26000种生化物质构成，而且其中绝大多数尚不为人知。

machine learning 机器学习

artificial intelligence 人工智能

propose v. 提出（观点）

diet n. 饮食

consist of 由……组成

at least 至少

the vast majority 绝大多数

¹⁴ This shows that we have some way to travel before achieving the first objective of systems thinking—which, in this example, is to identify more constituent parts of the nutrition system.

这表明，要实现系统思维的第一个目标，我们还有一段路要走；在这个例子中，这一目标就是找出营养系统中更多的组成部分。

have some way to travel 还有一段路要走 objective n. 目标

identify v. 识别；找出 constituent adj. 组成的；构成的

¹⁵ A systems **approach** to creating change is also **built on** the **assumption** that everyone in the system has **equal** **power**.

推动变革的系统方法还建立在这样一个假设之上：系统中的每个人都拥有平等的权力。

approach n. 方法

be built on 建立在……基础上

assumption n. 假设

equal adj. 平等的

power n. 权力

¹⁶ But as some researchers find, the food system is not an equal one.

但正如一些研究人员发现的那样，食物系统并不平等。

researcher n. 研究人员

equal adj. 平等的

¹⁷ A good way to **redress** (修正) such **power imbalance** is for more **universities** to do what Crystal did and teach students how to think using a systems **approach**.

要纠正这种权力失衡，一个好办法是让更多大学效仿克利斯特尔的做法，教学生如何运用系统方法进行思考。

redress v. 纠正；修正

power n. 权力

imbalance n. 不平衡；失衡

university n. 大学

approach n. 方法

¹⁸ More researchers, **policymakers** and **representatives** from the food **industry** must learn to **look beyond** their **direct** lines of **responsibility** and **adopt** a systems approach.

更多研究人员、政策制定者和食品行业的代表必须学会把目光投向自身直接职责范围之外，并采用系统方法。

policymaker n. 政策制定者

representative n. 代表

industry n. 行业；工业

look beyond 把目光投向……之外

direct adj. 直接的

responsibility n. 职责；责任

adopt v. 采用

¹⁹ Crystal knew that **visions** **alone** don't **produce** results, but **concluded** that “we'll never produce results that we can't **envision**”.

克利斯特尔明白，仅有愿景并不能带来成果，但她得出的结论是：“我们永远无法取得自己无法想象的成果。”

vision n. 愿景；设想

alone adv. 仅仅；单单

produce v. 产生；带来

conclude v. 得出结论

envision v. 设想；展望